Edward J. Carter

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Core Skills

Data & AI Platform Engineering: Distributed data platforms, ML pipeline integration, feature and inference data systems

Processing & Compute: Apache Spark (PySpark), large-scale data processing, batch and streaming integration

Vector & Retrieval Systems: Qdrant, pgvector, embedding pipelines, hybrid retrieval (RAG patterns)

Streaming & Event Systems: Apache Kafka, real-time data pipelines, event-driven architectures

Data Quality & Governance: Validation frameworks, anomaly detection, schema enforcement

Platform Infrastructure: Kubernetes, containerized ML environments, Databricks

Data Storage: PostgreSQL, NoSQL systems, hybrid structured/unstructured data systems

Languages: Python, SQL

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Professional Experience

Senior Data Engineer (Platform & Data Engineering) 12/2023 – Present
Ennoble First (SAFFIRE/CHINOOK)

- **Own and evolve enterprise data platform capabilities** supporting distributed processing, governed data access, and mission-critical analytics across engineering and operational teams
- **Define and implement data engineering standards** (ingestion, transformation, validation, lifecycle management), improving reliability and reducing rework across multiple teams and pipelines
- **Lead adoption of medallion-based architecture patterns** (Bronze/Silver/Gold), enabling consistent data lifecycle management and scalable, self-service data consumption
- **Establish data quality and validation frameworks**, improving trust, consistency, and usability of downstream datasets
- **Drive cross-functional alignment** across engineering, architecture, and operational stakeholders to translate mission requirements into scalable platform capabilities
- Improve accessibility and timeliness of enterprise data, enabling more reliable and responsive decision-making
- Contribute to technical strategy and proposal efforts, shaping solution architectures aligned with mission and organizational priorities
- Develop reusable architectural patterns that standardize distributed system design and support long-term platform

evolution

Data Engineer 09/2023 – 12/2023

Los Alamos National Laboratory (Programming & Runtime Environment)

- **Designed and deployed secure, containerized compute environments** across classified and unclassified domains, enabling scalable data processing and experimentation workflows
- **Architected microservice-based systems** supporting HPC resource utilization and license management, improving system reliability and operational efficiency
- **Led Kubernetes-based platform deployment and standardization**, increasing team productivity and enabling consistent, reproducible infrastructure delivery
- Contributed to platform engineering practices improving system observability, scalability, and operational consistency

Software Engineer / Data Engineer 05/2021 – 12/2023

Army Software Factory (Army Futures Command)

- **Led development and optimization of large-scale data pipelines processing 500M+ records**, reducing runtime from 24 hours to ∼26 minutes and enabling near real-time decision support
- Designed and delivered curated datasets into enterprise analytics platforms (Army Vantage), improving operational visibility and data accessibility across engineering and leadership stakeholders
- **Drove modernization of fragmented data systems into unified, scalable architectures**, increasing interoperability and enabling enterprise-wide data access
- Partnered across engineering, operations, and product stakeholders to align data platform capabilities with mission objectives and long-term priorities

Data Engineer 04/2021 – 09/2022

Texas ProTax

- Designed and implemented data integration workflows across SQL and NoSQL systems, enabling scalable pipelines for analytics and machine learning initiatives
- Delivered curated, analysis-ready datasets aligned with governance and modeling standards, improving data usability and accelerating downstream decision-making

Digital Intelligence Systems Master Gunner (DISMG) 03/2017 – 04/2021

United States Army

- **Led integration and modernization of enterprise intelligence data platforms** (DCGS-A CD2, Palantir-powered), aligning data architecture with mission-critical decision systems
- Directed cross-functional teams spanning analysts, engineers, and operational personnel to support data-driven mission execution
- **Owned reliability, accuracy, and availability of mission-critical data systems**, enabling consistent real-time decision support in operational environments
- Drove adoption of data governance standards and structured data practices across distributed teams

Intelligence Analyst / Data Analyst → Data Scientist → Data Architect 09/2002 – 04/2024

United States Army

- Progressed through increasingly senior roles, developing end-to-end ownership of the data lifecycle from ingestion and modeling to advanced analytics and decision support
- Designed scalable data architectures and standardized analytical workflows, enabling consistent and repeatable decision-making across distributed environments
- Served as a technical bridge between analysts, engineers, and leadership stakeholders, translating operational needs into scalable data solutions
- **Led and mentored personnel across multiple organizational levels**, contributing to workforce development, technical capability growth, and mission effectiveness

Representative Platform Systems

Large-Scale Medallion Data Platform *PySpark, Delta Lake, Parquet*

- *Architecture:* Medallion (Bronze/Silver/Gold) data platform with integrated data quality and incremental processing
- Designed and implemented an end-to-end data platform standardizing ingestion, transformation, and consumption across analytical workloads
- Established data quality validation and investigation patterns, improving trust and usability of downstream datasets
- Optimized transformation and partitioning strategies to support scalable analytics and machine learning workloads
- **Impact:** Reduced friction in accessing analysis-ready data and improved consistency of downstream datasets, enabling more reliable and timely decision-making

Hybrid Streaming & Batch Data Platform *Kafka, Spark, Python, Kubernetes*

- *Architecture:* Hybrid platform combining event-driven streaming with batch processing layers
- Designed ingestion and processing patterns supporting scalable, modular data transformation across real-time and batch workloads
- Implemented incremental processing and validation strategies aligned to modern data platform design
- Enabled consistent data flow patterns supporting downstream analytics and operational systems
- **Impact:** Reduced data latency and improved consistency between real-time and batch systems, enabling more timely and reliable downstream consumption

Geospatial Data Platform (DoD Context) *Databricks, Apache Sedona, H3, AWS*

- *Architecture:* Distributed geospatial data platform leveraging medallion modeling and spatial indexing (H3)
- Designed scalable geospatial processing workflows supporting large-scale spatial joins and query workloads
- Integrated spatial indexing and geospatial data models to enable high-performance spatial analytics
- Enabled map-based analytical workflows supporting operational decision-making at scale
- **Impact:** Improved performance and accessibility of geospatial analysis, enabling faster exploration and more responsive operational insights

Education & Certifications

- M.S. Data Engineering, Auburn University (Expected 2027)
- B.S. Database Management / Data Analytics, Western Governors University
- ITILv4 — CIW-Web Foundations — CIW-Data Analyst — CompTIA Project+